METHODOLOGY RECAP · MODULE M1

Data quality assessment

What the module does · How to read its outputs

WHAT IT DOES

Before anyone trusts a number, this module checks the data. It reads every facility's monthly reports and flags where the data looks shaky — so you know how much to rely on each indicator.

WHAT IT DOES NOT DO

It changes nothing. It only **measures** quality and shows you where the problems are. Fixing them is the next module's job.

THE THREE CHECKS

Outliers — values that look too high to be real

Completeness — months where a facility didn't report

Consistency — related numbers that don't line up

How to read the outputs

Every table in this module uses the same **traffic light**:

- **Green** — meets the quality standard
- **Gold** — borderline; check it
- **Red** — falls short; this data needs attention

What's behind each box. Every check works the same way underneath. Take one facility, one month, one indicator — that single report either passes the check or it doesn't. That's **one test**. Each coloured box then gathers all those tests for an area and shows the **share that passed** (for outliers, the share that *failed*). So a box really answers: *of all the reports behind it, how many were OK?*

Read each table closely — the colours guide your eye, but the number in every box matters: it tells you **which indicator** and **which area** you can rely on, and where you can't.

Each check looks at the data from a different angle — are the values realistic (outliers), are the reports arriving (completeness), and do related numbers agree (consistency). Together they tell you how far to trust each indicator.

> ***DQA measures quality — it does not change the data. Read it before you trust any trend; the next module repairs the problems it finds.***

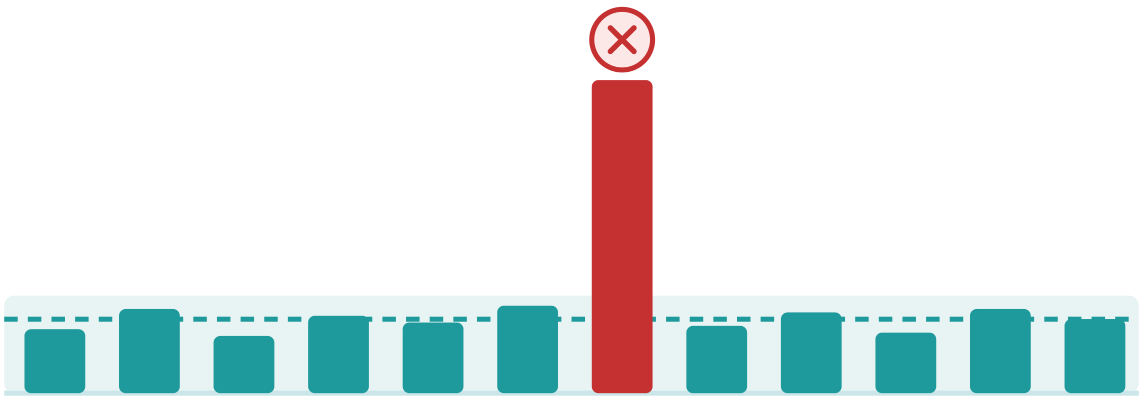
CHECK 1 OF 3 · OUTLIERS

Outliers — "is any month suspiciously high?"

How we find them. For each facility, FASTR learns what a normal month looks like for an indicator, then flags months that break the pattern in one of two ways:

- **A value far above normal.** Say a clinic usually reports 40–60 first antenatal visits a month, then one month shows 900. That is more than **10×** the facility's normal month-to-month variation, so it's flagged. (That variation is measured with the *median absolute deviation* — a robust average that one extreme month cannot distort.)
- **A single month dominates the year.** If one month holds more than **80%** of everything a facility reported for an indicator over the past 12 months, it's flagged — the typical sign of a whole year's total recorded in a single month.

Only indicators averaging more than 100 a month are checked, so small clinics aren't flagged for normal ups and downs.



CHECK 1 OF 3 · THE OUTPUT

Outliers — reading the table

Outliers

Percentage of facility-months that are outliers, May 2024 to Apr 2025

	Antenatal care 1	Antenatal care 4	Institutional delivery	Postnatal care 1 (newborns)	Postnatal care 1 (mothers)	BCG vaccine	Penta vaccine 1	Penta vaccine 3	Outpatient visit
Region 001	0.6%	0.2%	0.2%	1.9%	1.2%	0.6%	0.9%	0.6%	0.9%
Region 002	0.6%	0.0%	1.2%	1.0%	1.8%	0.2%	0.2%	0.1%	2.7%
Region 003	0.5%	0.4%	0.1%	0.0%	0.1%	0.1%	0.4%	0.6%	1.4%
Region 004	0.4%	0.0%	0.0%	0.8%	0.0%	0.5%	0.8%	0.9%	0.1%
Region 005	0.5%	0.9%	0.5%	0.8%	0.5%	0.6%	0.8%	0.3%	3.3%
Region 006	0.4%	0.0%	0.1%	0.3%	0.1%	1.3%	2.3%	2.5%	0.6%

3% or above
1% to 2%
Below 1%

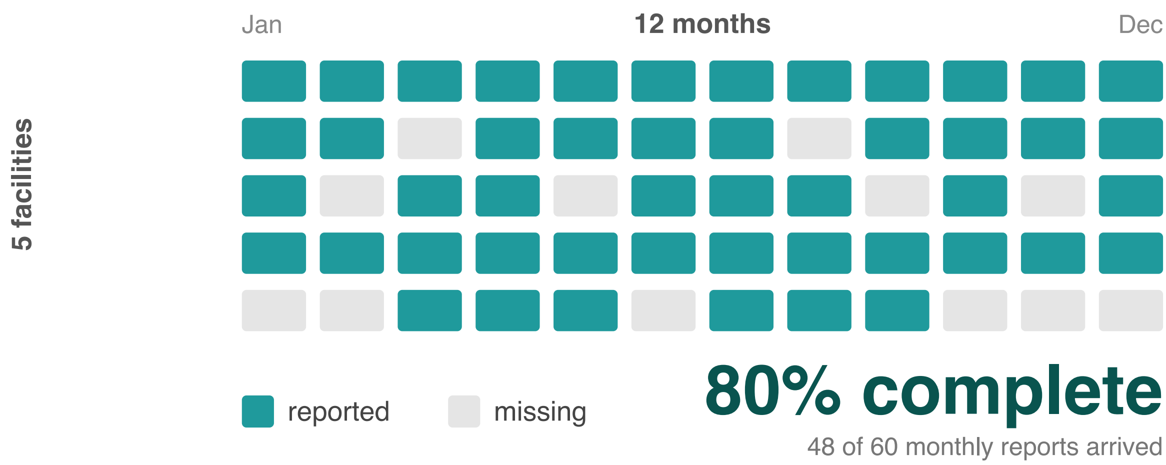
Outliers are reports which are suspiciously high compared to the usual volume reported by the facility in other months. Outliers are identified by assessing the within-facility variation in monthly reporting for each indicator. Outliers are defined observations which are greater than 10 times the median absolute deviation (MAD) from the monthly median value for the indicator in each time period, OR a value for which the proportional contribution in volume for a facility, indicator, and time period is greater than 80%. Outliers are only identified for indicators where the volume is greater than or equal to the median, the volume is not missing, and the average volume is greater than 100.

- **What each cell is** — pick a region (a row) and an indicator (a column). The number is how often that indicator's monthly reports looked **too high to be real** in that region. 0.5% means it almost never happened; 3% means about 1 report in every 33
- **How to read it** — **green is good** (hardly any suspicious months); **red means a lot**. A whole **red row** means that region enters careless numbers across many indicators. A whole **red column** means that one indicator is hard to report correctly everywhere

> *Worked example — in the table above, find **Region 005 → Outpatient visit: 3.3%, shown red.** That means about 1 of every 30 monthly outpatient reports in Region 005 was flagged as too high — likely a clinic entering a lump sum instead of one month. The rest of Region 005's row is green, so it's that one indicator that needs a look, not the whole region. FASTR repairs these in the next module.*

Completeness — "did facilities actually report?"

How we measure it. FASTR first works out the window when a facility was actually active for an indicator — from its first report to its last, setting aside long stretches (6+ months) right at the start or end, when it clearly wasn't open yet or had stopped reporting. Within that active window it counts how many months carry a number. A facility active for 12 months that reported in only 9 is **75% complete**. A blank counts as missing — FASTR can't tell "no service happened" from "nobody filed the report", so it treats both as a gap.



CHECK 2 OF 3 · THE OUTPUT

Completeness — reading the table

Indicator Completeness

Percentage of facility-months with complete data, Jan 2022 to Apr 2025

	Antenatal care 1	Antenatal care 4	Institutional delivery	Postnatal care 1 (newborns)	Postnatal care 1 (mothers)	BCG vaccine	Penta vaccine 1	Penta vaccine 3	Outpatient visit
District 001	93.9%	90.3%	83.3%	70.8%	82.8%	92.3%	91.9%	90.4%	94.9%
District 002	88.5%	85.9%	82.0%	64.2%	79.8%	90.8%	88.6%	88.1%	88.4%
District 003	89.8%	87.0%	85.4%	53.4%	78.5%	86.2%	86.3%	85.5%	88.3%
District 004	90.7%	82.6%	84.5%	67.6%	82.1%	89.1%	90.4%	89.7%	90.4%
District 005	89.7%	84.0%	81.3%	60.4%	80.9%	84.3%	84.2%	83.6%	89.5%
District 006	85.5%	79.4%	74.5%	56.8%	74.5%	85.3%	82.9%	79.2%	85.6%
District 007	97.2%	93.8%	91.8%	73.5%	90.8%	91.3%	95.2%	87.9%	97.0%
District 008	93.1%	87.6%	85.5%	44.6%	71.8%	79.9%	87.2%	87.5%	95.0%
District 009	95.7%	75.8%	85.0%	34.9%	77.6%	83.5%	92.2%	78.1%	95.9%
District 010	99.2%	99.6%	95.8%	98.4%	94.2%	94.6%	93.2%	88.7%	100.0%
District 011	98.0%	95.2%	94.4%	66.6%	91.7%	96.1%	95.4%	91.8%	98.2%
District 012	92.6%	86.4%	83.7%	64.5%	84.7%	84.7%	87.2%	86.7%	92.6%
District 013	93.4%	90.8%	90.7%	77.1%	88.2%	90.3%	90.8%	87.7%	93.1%
District 014	88.8%	79.1%	81.2%	77.9%	81.2%	90.0%	89.8%	86.3%	91.2%
District 015	93.0%	88.2%	85.3%	63.9%	83.1%	86.7%	88.8%	88.7%	93.1%
District 016	96.0%	85.1%	92.0%	65.7%	91.6%	96.1%	95.5%	88.6%	96.8%
District 017	90.1%	84.8%	90.4%	53.4%	83.0%	72.7%	79.3%	76.0%	89.5%
District 018	97.1%	95.9%	93.2%	79.8%	92.3%	95.4%	95.8%	92.9%	97.9%

90% or above
80% to 89%
Below 80%

Higher completeness improves the reliability of the data, especially when completeness is stable over time. Completeness is defined as the percentage of reporting facilities each month out of the total number of facilities expected to report. A facility is expected to report if it has reported any volume for each indicator anytime within a year. A high completeness does not indicate that the HMIS is representative of all service delivery in the country, as some services may not be delivered in facilities, or some facilities may not report.

- **What each cell is** — pick a district (a row) and an indicator (a column). The number is how many of the months that district *should* have reported actually arrived. 92% means 92 of every 100 expected reports came in
- **How to read it** — **green is good** (almost all reports arrived); **red means many are missing**. A whole **red column** is an indicator hardly anyone reports (maybe new, or unclear how). A whole **red row** is a district that reports weakly across the board

Worked example — in the table above, find District 005 → Antenatal care 1: 69.7%, shown red. *Only about 7 of every 10 ANC1 reports that district should have sent actually arrived. A trend or coverage line built on that cell is standing on thin data — read it with caution.*

CHECK 3 OF 3 · CONSISTENCY

Consistency — "do related numbers make sense together?"

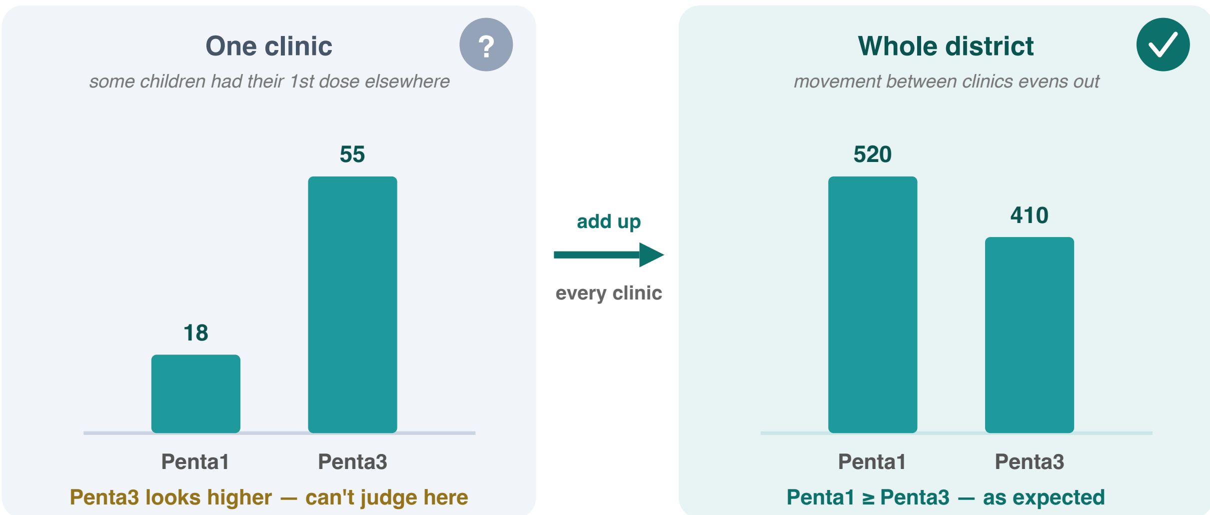
How we check it. Related indicators must keep a sensible order: more children get the 1st vaccine dose than the 3rd (**Penta1 ≥ Penta3**), more women a 1st antenatal visit than a 4th (**ANC1 ≥ ANC4**), and deliveries roughly match BCG doses (the jab given at birth).

At a **single clinic** these can break for innocent reasons — the numbers are small, and a child may get one dose at an outreach session and the next at a clinic. So FASTR adds up the **whole district** before checking; there, the comings and goings even out and the relationship should hold.

The check itself is just a **ratio**: FASTR compares each pair across the district — Penta1 against Penta3, ANC1 against ANC4 — and flags any that falls outside the plausible range. That's the whole point: a district reporting more 3rd doses than 1st is impossible in real life (no child gets a 3rd dose without a 1st), so the ratio flags it as an error.

The rule: Penta1 ≥ Penta3

More children get the 1st vaccine dose than the 3rd (some drop out before finishing)



CHECK 3 OF 3 · THE OUTPUT

Consistency — reading the table

Internal consistency

Percentage of sub-national areas meeting consistency benchmarks, May 2024 to Apr 2025

	ANC1 is larger than ANC4	Delivery is approximately equal to BCG	Penta1 is larger than Penta3
Region 001	100.0%	0.0%	97.2%
Region 002	100.0%	0.0%	100.0%
Region 003	100.0%	45.8%	83.3%
Region 004	100.0%	37.5%	87.5%
Region 005	100.0%	0.0%	87.5%
Region 006	100.0%	49.0%	84.4%

90% or above
80% to 89%
Below 80%

Internal consistency assesses the plausibility of reported data based on related indicators. Consistency metrics are approximate - depending on timing and seasonality, indicator definitions, and the nature of service delivery and reporting, values may be expected to sit outside plausible ranges. Indicators which are similar are expected to have roughy the same volume over the year (within a 30% margin). The data in this analysis is adjusted for outliers.

- **What each cell is** — pick a region (a row) and a pair of related indicators (a column). The number is the share of that region's **districts** where the two numbers line up the way they must
- **How to read it — green is good** (the rule holds in almost every district); **red means it's broken in most**. The three rules: more 1st antenatal visits than 4th (**ANC1 ≥ ANC4**), more 1st than 3rd vaccine doses (**Penta1 ≥ Penta3**), and deliveries ≈ BCG doses

> ***Worked example — Region 002 → "Delivery ≈ BCG": 0.0%, red** means that in none of Region 002's districts do delivery and BCG numbers match. Often that's because the two are recorded in different places, not a true error — so treat this pair as a gentler warning than the antenatal and vaccine ones, which should hold tightly.*

THE SUMMARY

The DQA score — one number per region, per year

How it's built. Look at one facility in a single month. It counts as **clean** only when every check passes there: the core indicators (outpatient visits, Penta1, ANC1) have no missing reports and no outliers, and the related pairs (Penta1/Penta3, ANC1/ANC4) line up. The DQA score is just the share of those facility-months that come out clean — so **84% means 84 of every 100 were clean and trustworthy.**

Overall DQA score

Percentage of facility-months with adequate data quality over time

	2022	2023	2024	2025
Region 001	60.8%	76.6%	76.4%	84.4%
Region 002	55.3%	61.2%	58.2%	52.0%
Region 003	69.3%	73.4%	60.6%	47.0%
Region 004	54.6%	70.7%	70.0%	84.9%
Region 005	57.9%	69.5%	56.6%	55.4%
Region 006	74.0%	84.7%	72.2%	71.7%

80% or above
70% to 79%
Below 70%

Adequate data quality is defined as: 1) No missing data or outliers for OPD, Penta1, and ANC1, where available 2) Consistent reporting between Penta1/Penta3 and ANC1/ANC4.

- **How to read it** — **green is good** (most facility-months are clean). Read **left to right** to see whether a region is improving year on year, and **top to bottom** to compare regions in one year

> ***Worked example — Region 001 climbs 60.8% → 84.4% from 2022 to 2025 (red to green): its data became steadily more trustworthy. Region 003 slips to 47.0% by 2025 (red) — that's where data quality needs attention first. Then move to the adjustment module, which repairs the problems found here.***